Customer Churn Prediction & Retention Strategy

Business Problem

Background

Customer churn is a critical challenge in the telecom industry. When customers discontinue their service, the company loses recurring revenue and must spend additional resources to acquire new customers. Since acquiring a new customer is more expensive than retaining an existing one, reducing churn is strategically important for long-term profitability.

Problem Statement

The telecom company is experiencing customer attrition, leading to revenue loss and reduced customer lifetime value. The business needs to:

- Measure the extent of churn
- Identify high-risk customer segments
- Understand key churn drivers
- Build a predictive model to identify likely churners
- Develop data-driven retention strategies

Dataset Overview

The dataset contains:

- 7,043 customer records
- 33 original features
- Customer demographics
- Service subscriptions
- Billing details
- Churn indicators
- Churn reasons

After cleaning:

- 30 features retained
- Missing values handled
- No duplicate records
- Target variable: `churn_value` (1 = churn, 0 = retained)

SQL Analysis Findings

Initial SQL exploration revealed:

Overall Churn Rate

- ~26.5% of customers churned
- Significant revenue impact from churned customers

High-Risk Segments

Segment	**Churn Rate**
Month-to-month contract	42.7%
Senior citizens	41.6%
Electronic check payment	45%
Fiber optic users	41.9%

Key Observations

- Short-term contracts have much higher churn
- Senior citizens are significantly more likely to churn
- Electronic check users churn disproportionately
- Fiber optic customers show elevated churn rates

These patterns suggested both behavioral and pricing-related churn drivers.

Exploratory Data Analysis (Python)

Visual analysis confirmed:

◇ Tenure

Customers in the first 12 months showed the highest churn concentration.

◇ Pricing

Higher monthly charges were associated with increased churn probability.

◇ Service Bundling

Customers without:

- Tech support
- Online security
- Device protection
  were more likely to churn.

◇ Payment Behavior

Electronic check users had significantly higher churn rates.

The EDA reinforced the SQL findings and revealed deeper service-related patterns.

Machine Learning Modeling

Two models were built and compared:

Logistic Regression

Metric	Value
Accuracy	80%
Precision	0.64
Recall	0.57
F1 Score	0.60

Random Forest

Metric	Value
Accuracy	79%
Precision	0.62
Recall	0.52
F1 Score	0.57

Final Model Selection

Logistic Regression was selected due to:

- Higher recall (better churn detection)
- Slightly higher F1 score
- Simpler, more interpretable structure

Top Churn Drivers (ML Feature Importance)

Random Forest feature importance identified:

1. Total charges
2. Tenure months
3. Monthly charges
4. Fiber optic internet
5. Electronic check payment
6. Dependents
7. Contract type (Two-year)
8. Online security

These aligned strongly with SQL and EDA insights, increasing confidence in the findings.

Power BI Dashboard

A three-page interactive dashboard was developed:

Page 1 — Executive Overview

- Total customers
- Total churned
- Churn rate
- Revenue lost
- Churn by contract
- Churn by internet service

Page 2 — Customer Segmentation

- Churn by payment method
- Churn by senior citizen
- Churn by tenure group
- Churn by monthly charge group

Page 3 — Service & Behavioral Drivers

- Churn by tech support
- Churn by online security
- Churn by device protection
- Top churn reasons

The dashboard allows interactive filtering and executive-level analysis.

Business Insights

The strongest churn drivers identified were:

- Low commitment (month-to-month contracts)
- Early-stage customers
- High monthly charges
- Fiber optic service users
- Electronic check payment method
- Lack of bundled services
- Senior citizen segment

Churn is primarily driven by:

- Pricing sensitivity
- Low service engagement
- Competitive pressure
- Service dissatisfaction

Business Recommendations

1. Promote Long-Term Contracts

Offer discounts for 1–2 year plans.

2. Improve Early Customer Engagement

Implement onboarding and retention campaigns for first-year customers.

3. Review Fiber Pricing & Service Quality

Investigate dissatisfaction among fiber customers.

4. Encourage Auto-Pay Enrollment

Offer incentives for automatic payment methods.

5. Bundle Value-Added Services

Promote online security and tech support add-ons.

6. Create Senior-Focused Retention Plans

Simplify plans and provide enhanced support.

Expected Business Impact

If churn is reduced by even 5%:

- Significant monthly revenue can be retained
- Customer lifetime value increases
- Retention marketing becomes more efficient
- Customer acquisition costs decrease

- # Conclusion
- This end-to-end churn analysis combined SQL, Python, machine learning, and Power BI to identify high-risk segments and churn drivers. Logistic Regression provided a reliable predictive baseline, while Random Forest validated key features.
- The analysis revealed that churn is driven primarily by contract structure, tenure, pricing, service engagement, and payment behavior. Targeted retention strategies focused on these areas can substantially reduce churn and protect revenue.

Project Tools Used

- PostgreSQL (SQL Analysis)
- Python (Pandas, Seaborn, Scikit-learn)
- Power BI (Dashboard)
- Joblib (Model Saving)